THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET A

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Which solid has all faces as congruent equilateral triangles? 1

(a) square pyramid **(b)** cube **(c)** prism **(d)** regular tetrahedron

2. A solid has 8 faces and 12 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 22 **(b)** 18 **(c)** 16 **(d)** 20

3. Each side of a Koch curve is replaced at each stage by how many segments? 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 5

4. The median of 30, 35, 40, 45, 50, 60, 65 is: 1

(a) 44 **(b)** 49.4286 **(c)** 46 **(d)** 45

5. The mean of 4 numbers is 50. 3 of them are 40, 50, 60. The remaining number is: 1

(a) 50 **(b)** 55 **(c)** 52 **(d)** 48

6. The mean of the data 5, 6, 6, 9, 11, 13, 14 is: 1

(a) 11.1429 **(b)** 9.14286 **(c)** 7.14286 **(d)** 10

7. Think of a number n. Add 5, multiply by 2, then subtract 10. The result is: 1

(a) $2n$ **(b)** $2n + 5$ **(c)** $5n$ **(d)** $n + 5$

8. For any integer n, $(n + 2)^2 - (n - 2)^2$ equals: 1

(a) $4n$ **(b)** 8 **(c)** $8n$ **(d)** 4

9. A number is divisible by 9 if and only if the sum of its digits is divisible by: 1

(a) 6 **(b)** 12 **(c)** 9 **(d)** 3

10. The area of a rhombus with diagonals 16 cm and 12 cm is: 1

(a) 192 cm² **(b)** 28 cm² **(c)** 48 cm² **(d)** 96 cm²

11. The area of a rectangle of length 15 cm and breadth 8 cm is: 1

(a) 120 cm² **(b)** 128 cm² **(c)** 46 cm² **(d)** 23 cm²

12. A rectangle has area 200 cm² and length 25 cm. Its breadth is: 1

(a) 175 cm **(b)** 8 cm **(c)** 7 cm **(d)** 10 cm

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Verify Euler's formula for a triangular prism. 2

14. State when a line graph is suitable. 2

15. Find the mean of 12, 18, 24, 30, 36. 2

16. The bottom row of a number pyramid is 3, 5, 4. Find the top block. 2

17. How many 25 cm × 20 cm tiles cover a floor of 5 m by 4 m? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Find the faces, edges and vertices of a hexagonal pyramid and verify Euler's formula. 3

19. Find the mean and the median of: 8, 8, 10, 12, 14, 18. Which is greater? 3

20. Use a number pyramid with bottom row 5, 8, 3: find all rows and the top, and write the top in terms of the bottom numbers. 3

21. Find the area of a trapezium with parallel sides 12 cm and 20 cm and distance between them 8 cm. Also find the area of an equal-height rectangle with width 20 cm. 3

22. A rectangular plot is 32 m by 15 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

OR

Find the area of a right triangle with hypotenuse 25 cm and one leg 7 cm. 3

23. A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 4, find the number of shaded triangles and the side of each. 3

OR

A cube has edge 2 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $4 \times 2 \times 2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Show that the mean of the first n natural numbers is $(n + 1)/2$. 5
(ii) Hence find the mean of the first 50.

OR

(i) Define mean and median. 5
(ii) Find both for 14, 20, 12, 16, 18, 40.
(iii) Which is a better measure of the typical value?

25. **Case study:** A rhombus-shaped tile has diagonals 24 cm and 18 cm. 5
(i) Find the area.
(ii) Find the side.
(iii) How many tiles to cover 5.4 m² (ignore waste)?

OR

(i) Find the area of a quadrilateral with a diagonal of 20 cm and offsets 7 cm and 9 cm to the diagonal. 5
(ii) Name the method.

END OF QUESTION PAPER — Total: 50 Marks